

Sales Prediction & Customer Segmentation using Linear Regression & K-Means Clustering.

AIM:

To predict sales based on advertising budgets using Linear Regression & K-Means Clustering

PROCEDURE:

1. Load advertising.csv dataset
2. Select features (TV, Radio, Newspapers) & target (Sales)
3. Split data into training & testing sets.
4. Train Linear Regression, predict & calculate MSE
5. Plot actual vs predicted sales
6. Standardize features & apply K-Means clustering (3 clusters)
7. Visualize.

CODE:

```
import pandas as pd, seaborn as sns, matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

df = pd.read_csv("advertising.csv")
x, y = df[['TV', 'Radio', 'Newspaper']], df['Sales']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=0)
```

```
model = LinearRegression().fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

```
print("MSE:", mean_squared_error(y_test, y_pred))
```

```
sns.scatterplot(x=y_test, y=y_pred)
```

```
plt.xlabel("Actual Sales"); plt.ylabel("Predicted Sales")
```

```
plt.title("Linear Regression: Actual Vs Predicted")
```

```
plt.show()
```

```
scaled = StandardScaler().fit_transform(X)
```

```
df['cluster'] = KMeans(n_clusters=3, random_state=0).fit_predict(scaled)
```

```
sns.scatterplot(x=df['TV'], y=df['Sales'], hue=df['cluster'], palette='Set1')
```

```
plt.title("K-Means Clustering")
```

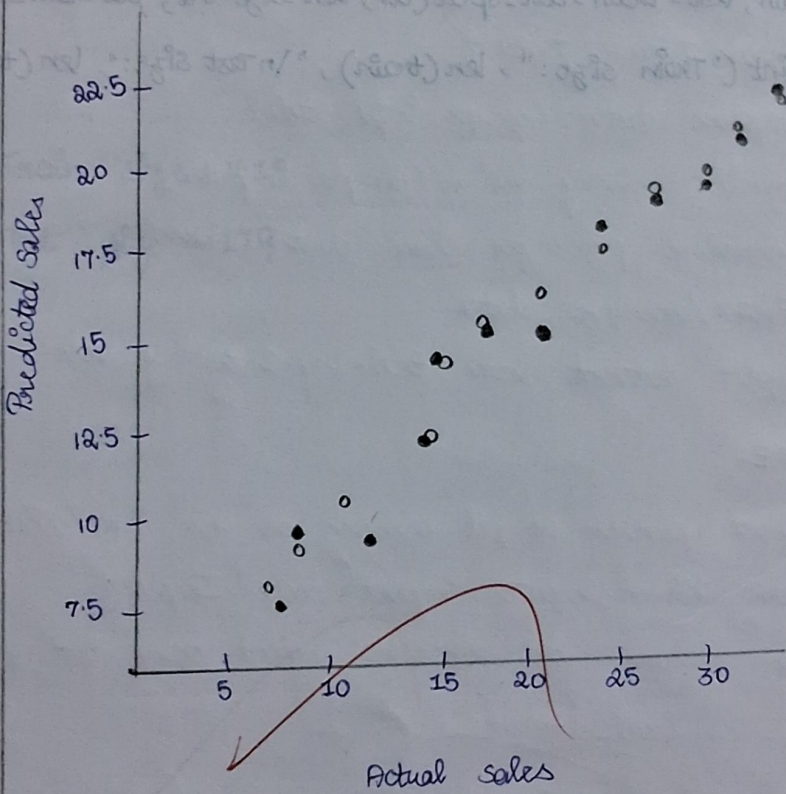
```
plt.show()
```

RESULT:

Thus a Linear Regression Model is Built & the Clustering is made into 3 distinct clusters, revealing clear patterns in TV advertising.

OUTPUT:

Linear Regression: Actual Vs Predicted



K-Means clustering: TV Budget Vs Sales

